



Mid Semester Examination Autumn 2024-2025

Class: **MTech**

Time: 2Hrs

Course Title: **Computer Vision**

F.M.: **30**

Course ID: **EC-6603**

**This question paper
contains 2 pages.**

Answer all questions.

Figures in the right-hand margin indicate marks.

1. (a) Which of the thresholding techniques is appropriate when image histogram has more than two modes?
(b) Discuss the effect of lowpass filtering and highpass filtering on images.
(c) An object is 16 cm wide and is imaged with a sensor of size 8mm(H) × 6mm(W) from a distance of 0.5m. What should be the required focal length of the lens used in the camera so that the object is properly focused?
(d) Describe Contrast Stretching.
(e) Distinguish between mean square error (MSE) and mean absolute error (MAE). 1×5
2. Apply histogram equalization operation on a 4-bit image having the histogram shown in the following table.

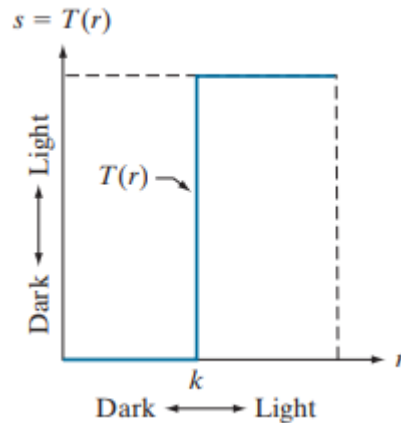
Gray levels	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Number of pixels	1	4	4	1	4	1	5	6	3	16	3	12	7	16	6	11

Table.1 Image grey level distribution

- Plot the histograms of the input and output images. Comment on the result. What will happen if we apply histogram equalization repetitively? 5
3. Find the normalized kernel for a 5×5 Low Pass Gaussian Filter with $\sigma = 1, 2, \text{ and } 3$ (three different cases). How does the filter behaviour vary with increasing standard deviation? 5
 - 4.. (a) What is Laplacian? Derive the formula for it and hence design a 3×3 mask for this operation. 2
(b) Why and when should a designer think of filtering in frequency domain instead of spatial-domain? Determine how much computational advantage he/she will have, if he/she does so. Describe the steps involved in frequency-domain filtering. 3
 5. In a character recognition application, text pages are reduced to binary using a thresholding transformation function of the form in Fig. 1. This is followed by a procedure that thins the characters until they become

strings of binary 1's on a background of 0's. Due to noise, binarization and thinning result in broken strings of characters with gaps ranging from 1 to 3 pixels. One way to "repair" the gaps is to run a smoothing kernel over the binary image to blur it, and thus create bridges of nonzero pixels between gaps.

Fig. 1



- (a) Give the (odd) size of the smallest box kernel capable of performing this task.
 - (b) After bridging the gaps, the image is thresholded to convert it back to binary form. For your answer in (a), what is the minimum value of the threshold required to accomplish this, without causing the segments to break up again?
6.
 - (a) A CCD TV camera is used to perform a long-term study by observing the same area 24 hours a day, for 30 days. Digital images are captured and transmitted to a central location every 5 minutes. The illumination of the scene changes from natural daylight to artificial lighting. At no time is the scene without illumination, so it is always possible to obtain an acceptable image. Because the range of illumination is such that it is always in the linear operating range of the camera, it is decided not to employ any compensating mechanisms on the camera itself. Rather, it is decided to use image processing techniques to post-process, and thus normalize, the images to the equivalent of constant illumination. Propose a method to do this. You are at liberty to use any method you wish, but state clearly all the assumptions you made in arriving at your design.
 - (b) An automobile manufacturer is automating the placement of certain components on the bumpers of a limited-edition line of sports cars. The components are color-coordinated, so the assembly robots need to know the color of each car in order to select the appropriate bumper component. Models come in only four colors: blue, green, red, and white. You are hired to propose a solution based on imaging. How would you solve the problem of determining the color of each car, keeping in mind that cost is the most important consideration in your choice of components?
 - (c) Find the memory requirement for storing a color(R,G,B) image of size 1024x2048 with each color channel quantized to 8 bits.

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